EVALUATING INTEGRALS

Splitting a fraction into standard forms

Expand, simplify, and separate before reaching for techniques.

01

ONE FRACTION — BEFORE

$$\int \frac{x^3 + 1}{x}\,dx$$

↓ `Split the fraction` — a single-term denominator: $(x^3 + 1)/x = x^3/x + 1/x$

02

SUM OF POWERS OF X

$$\int \left(x^2 + \frac{1}{x}\right) dx$$

↓ `Integrate term by term` — the standard forms $\int x^n\,dx$ and $\int 1/x\,dx$, by linearity

03

ANTIDERIVATIVE — AFTER

$$\frac{x^3}{3} + \ln|x| + C$$

▼ NOT TOP AND BOTTOM SEPARATELY